Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2017

Digital Technology

Assessment Unit AS 1

assessing

Approaches to System Development

[CODE]

SPECIMEN

MARK SCHEME

Version 2: 6 March 2017

AVAILABLE MARKS

1 **(a)** It was recognised that the way in which software is developed must be radically changed [1], because users' needs were not being met/ users' needs were being met [1]. This is to ensure software meets the user requirements [1]. Software is developed on time/not on time [1] and within budget/outside budget [1]. Quality process not being followed [1]. Effective advances in hardware and software were not being used in the production of software e.g. IDE [1]

Any **four** points [4]

(b) Plan the project's schedule [1]
Split the project up into tasks/subtasks [1]
Allocate resources such as personnel, Digital Technology to each task/subtask [1]
Identify risks to the project [1]
Monitor the progress of the project [1]

Any **two** specific duties [2]

(c) This is used to find out how data flows through the organisation [1]. It looks at how data is input [1], collated [1], processed [1] and reported [1]. Typical documents are data collection forms, invoices, orders, receipts [1]

Any **three** points [3]

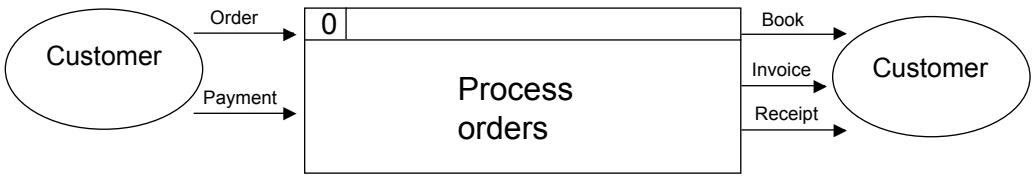
(d) Functional requirements specify what a software system must do. [1]

Example: when the shock level for a product falls below it's re-order level, a re-order request should automatically be generated. [1]

Non-functional requirements specify additional criteria by which the system will be judged. [1]

Example: the sale of an item at a POS terminal should be recorded in the database within x seconds. [1] [4]

(e)



[1] for 'Process orders' (or similar)
[1] for identifying Level 0
[1] for Customer as data source/destination
[1] for two data flows in
[1] for three data flows out
[1] for correct symbols [6]

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AVAILABLE MARKS

2 **(a)** Specification of data structures [1]
Design of user interfaces [1]
Design of queries and reports [1]
Identification of the data validation to be used [1]
Identification of the data verification to be used [1]

Any **three** activities [3]

(b) Diagrams [1] will be produced for each page of the website [1]. Each diagram will show the user interface of the page [1]. This will include user input/output [1] and the different navigation paths available to the user [1]

Any **four** points [4]

(c) The item (e.g. module or text field) to be tested [1]
The test data to be used [1]
The expected outcome [1]

Any **two** points [2]

(d) Data is taken from existing system [1]. Data converted into a suitable format [1]
Non-electronic data may have to be converted to electronic form [1]
File formats may need to be changed [1]
Non-electronic data is taken from existing system [1] is converted to electronic form [1]

Any **three** points [3]

(e) **Indicative content**
Candidates may discuss the following:

The method
The new system replaces the existing system in one go
The old system is discarded
This is usually done at a quiet processing period e.g. overnight

Advantage
This method is very efficient because there is no duplication of Digital Technology resources or personnel.

Disadvantage
There is a potential high risk with this method – if the new system fails in any way there is no backup system available

The context
The risk should be low since payroll is a standard business process and follows established steps which can be tested in advance, and payroll is an isolated function within a business and any risk will be confined to one business area.

Level of response	Marking criteria	Marks
Band 3: Excellent	The candidate accurately describes the changeover method. The candidate addresses both advantages and disadvantages of this approach and refers directly to the context of a new payroll system in a coherent manner. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]–[9]
Band 2: Good	The candidate describes the changeover method, addressing some advantages and disadvantages of this approach but does not apply this directly to the context of a new payroll system. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]–[6]
Band 1: Basic	The candidate refers to the changeover method and identifies an advantage or disadvantage. The candidate makes no reference to the new payroll system. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]–[3]
The response is not worthy of credit		[0]

AVAILABLE MARKS

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AVAILABLE MARKS

3 **(a)** Consists of a number of distinct stages e.g. analysis [1] stages are sequenced [1]
One stage must be completed before the next stage commences [1]
There is a deliverable at the end of each stage e.g. system specification [1]
If an error is found at any stage, an earlier stage may have to be revisited [1]

Any **four** points [4]

(b) A task/subtask might take too long [1]. This could mean that tasks/ subtasks depending on its completion may not be able to start as scheduled [1]. Task may exceed its budget [1]. This would lead to software not being developed until financial constraints are resolved [1]
A key member of the team might be unavailable when required [1]
This could mean that a task/subtask may not be able to start [1]

Any **four** points [4]

(c) A Gantt chart is similar to a bar chart [1]. It identifies each project activity [1] and its planned start time [1] and finishing time [1]. It shows the current progress of the project with a vertical bar [1]

Any **four** points [4]

(d) A model of the project is constructed showing each activity as a node [1], and the dependencies between the activities [1]
A mathematical algorithm is used to calculate the earliest and latest starting and finishing time for each activity [1]

The algorithm identifies the chain of activities critical to the project's schedule [1]

CPA identifies the most efficient way of ensuring the project's schedule is met with optimal use of resources [1]

Use of CPA requires the use of special software and expertise on the part of the project manager and may be unsuitable for small scale projects [1]

Any **four** points [4]

(e)

Label	Description
2	*The node number*
4	Earliest start day [1]
10	Latest finish day [1]
D	The name of the activity [1]
3	The duration of the activity [1]

[4]

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AVAILABLE MARKS

4 **(a)** Errors may not be identified until the system has been in use [1]
These errors must be debugged and the system retested [1] [2]

(b) The user requirements may change after the system has been in use for some time [1]. This may be because of internal changes [1] or because of external changes [1]. In this case, adaptive maintenance is required [1]

Any **three** points

The system may still meet the user requirements [1] but the performance of the system may require improving [1]. This could be to take advantage of new hardware and software to improve processing speeds [1]. In this case, perfective maintenance is required [1]

Any **three** points [6]

(c) It will describe how the organisation will continue to function after a natural disaster [1]. It will identify the main risks to the organisation from possible disasters [1]. It will identify critical data [1]. It will identify recovery method to be used [1]. It will identify critical data processing operations [1]. It will dentify key personnel [1]. It will ensure personnel continue to have access to the organisation's computer systems [1]. It will specify the procedures to be performed if a disaster occurs [1]. It may identify an alternative location where the organisation could operate until the threat/damage is over [1]

Any **four** points [4]

(d) **Indicative content**
Candidates may discuss the following:

Differential backup
Data files which have changed
…since the last full backup are copied.
The regular backup procedure consists of a differential backup.
Full backups are done less frequently.

Advantage
This is an efficient process as differential backup do not take as long.
The amount of disk space used by differential images is significantly less than that of a full image. Only two image files are required to restore the system.

Disadvantage
If the system fails and data has to be recovered
…the recovery process is more complicated.
The last full backup has to be restored first
…followed by all subsequent differential backups.
The longer the time betweem the full and the differential, the larger the differential image file will be and the longer it will take to create.

Level of response	Marking criteria	Marks
Band 3: Excellent	The candidate accurately describes how differential backups are used in this combination of backup methods, clearly articulating both the advantages and disadvantages of this method. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]–[9]
Band 2: Good	The candidate describes this combination of backup methods, and sets out either the advantages or disadvantages of this method. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]–[6]
Band 1: Basic	The candidate demonstrates an awareness of a differential backup, and refers to an advantage or disadvantage of this method. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]–[3]
The response is not worthy of credit		[0]

AVAILABLE MARKS

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AVAILABLE MARKS

5 **(a)** A smart programmer editor [1]
Run/debug options [1]
Automatic production of documentation [1]
Project management features [1]

Any **two** points [2]

(b) A sequence of instructions/code [1]
...is repeated a number of times [1]
A variable will be used to control a loop [1]
This variable will be initialised [1]
...and incremented each time the loop is executed [1]
The loop will be performed until the value of the variable reaches 100 [1]

Any **four** points [4]

(c) **Object**
A self-contained component that incorporates data/properties [1] and functions/methods [1]

Inheritance
The mechanism whereby the properties and methods of one class [1] can be automatically copied to another class [1] [4]

(d) **Indicative content**
Candidates may discuss the following:

The approach
Advantages
More accurate modelling – the object oriented approach models the real world more realistically than traditional methods which focus on data and processing.
Improved productivity due to code reusability – when a new object is created, it will automatically inherit the data attributes and characteristics of any class from which it is derived.
The primary goal of object-orientated development is the assurance that the system will enjoy a longer life while having far smaller maintenence costs. Because most of the processes within the system are encapsulated, the behaviours may be reused and incorporated into new behaviours.

Disadvantages
The object oriented approach is not suited for all applications.
It has been widely adopted in CAD/CAM and engineering design systems but it has not been as successful for either corporate systems or many 'run of the mill' information systems applications such as payroll and accounting.
Object-orientated Development is not yet completely accepted by major vendors. While there is a general acceptance of object-orientated systems in the commercial marketplace, organisations do not have the time to train their staffs in object-orientated metheods.

Level of response	Marking criteria	Marks
Band 3: Excellent	The candidate accurately describes how the use of the OO approach can improve software quality. The candidate clearly articulates both the advantages and disadvantages of this approach. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]–[9]
Band 2: Good	The candidate describes the OO approach but doesn't refer to software quality. The candidate describes the advantages or disadvantages of this approach. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]–[6]
Band 1: Basic	The candidate makes some reference to the OO and doesn't refer to software quality. The candidate refers to an advantage or disadvantage of this approach. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]–[3]
The response is not worthy of credit		[0]

Total

AVAILABLE MARKS

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100